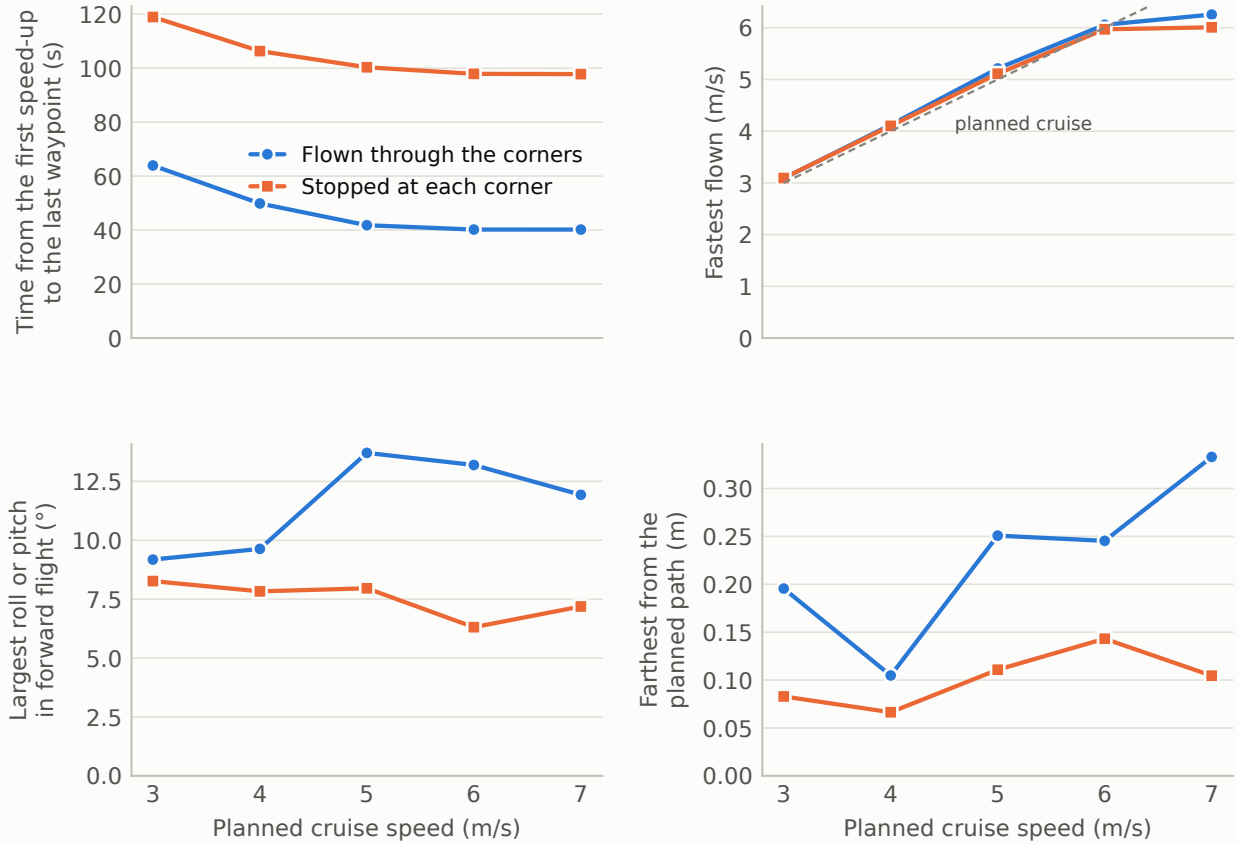
Flying through the corners halves the square (42 s against 100 s at 5 m/s); above 5 m/s it gains little

The 50 m square, no wind, each run alone. The corners of 15 m radius are taken at most sqrt(1.7 m/s² × 15 m) = 5 m/s, so a faster cruise is only reached on the first and last leg, and the middle legs speed up and slow down without a cruise.



PX4 v1.17.0 SITL with Gazebo and assumed vehicle coefficients; simulation, not flight data.